Overview on Hiding Data in Microsoft Office Documents

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***Abstract*—Steganography is the science of hiding information in various types of data. Some of the most popular formats for Microsoft Office electronic documents are docx, xlsx, and pptx. The paper analyzes the structure of documents according to the OOXML standard. Algorithms for embedding data in electronic documents such as docx, xlsx and pptx, using the functionality of Microsoft office, as well as corresponding to a zip archive, are considered. There are six types of data attachments: zip comments, restricted zip fields, file embedding, additional Xml fields, hidden text, and tracking text. The stability of the data algorithm and the main advantages and disadvantages are assessed.**

Keywords—steganography, analysis, OOXML, data hiding

I. Introduction

The internet and IT-technologies in general are improving, developing and spreading across the world year by year. Due to numerous new functions and upgraded technologies and systems, communication process management and control are becoming more and more challenging. For this reason, the demand for information protection has also increased. Leakage of confidential information represents one of the threats related to information security. Organizations, where information of great value is processed and stored, have leakage prevention systems. The chance of using steganographic methods is particularly high in such cases.

Steganography is the science that allows hiding communicated data inside a certain container to conceal the fact of communication. Steganography is used to protect copyright [1], secret media transmission [2]. Due to the widespread use of electronic document management, Microsoft Office documents are great in file containers. This article reviews different applications of steganographic methods using the specifics of document functions and the document's file structure for hidden information in electronic Microsoft documents using .docx, .xlsx, and .pptx formats. The new structure of MS Office 2007-2019 documents, which is based on the Office Open Extensible Markup Language (OOXML) format, can be perfectly used as a container for communicating secret information since its format has become open and described in detail in the ISO/IEC 29500 [3] standard: Office Open XML File Formats.

This makes such documents, objects for the introduction of secret messages - for further leakage of personal data [4] or confidential data [5].Recent cases of using steganography in MS Office documents during large hacker attacks [6, 7]. These data leakage detection (DLP) systems are not able to detect data hidden in steganographic containers and allow you to add them to their composite functions independently. Therefore, cybersecurity officers need to independently develop utilities to detect steganographic containers..

This article describes possible and additional information hiding methods for documents in the OOXML format, based on an in-depth study of this format. In addition, an assessment of each of the considered methods is given, the conclusions of which will expand the functionality of revealing hidden information contained in MS Office documents.

II. File Structure

MS Office-format files represent a zip-archive containing [8] XML for office documents including text documents, electronic tables, graphs, diagrams, figures and other graphic materials. An archive is a binary file containing an array of bytes. Also, every file in the archive has a name, attributes (e.g. file modification time) and other properties displayed in the file system related to any operating system. Therefore, in addition to zipped data, the name of the zipped file, its attributes and other private information are stored in every ZIP-archive. This private information is arranged in a strictly defined order and has regular structure.

A ZIP-file consists of three areas:

- Archived/Unarchived data
- Central directory
- Description of central directory

A Zip-archive can be regarded, in more detail, as a block sequence:

- Local File Header (LFH): this data block contains basic information on the zipped file including the file size before and after zipping, file modification time, hash function, CRC-32 and local file name pointer.
- File Name is a sequence of bytes of random length that generate the name of the zipped file.
- File Data is the zipped file content in the form of a byte array of random length. If the file is empty or represents a directory, this array will not be used and the file/directory name is immediately followed by the LFH title describing the next file.
- The Central Directory Header (CDH) contains extensive data representation in LFH. In addition to data comprised in LFH, the CDH contains file attributes, local reference to LFH structure and certain other information, which is generally not used.
- End of central directory record (EOCD): this structure is provided as a single copy in each archive and recorded at the very end of the archive. The most interesting data that it contains includes the number of records made in the archive (or the number of files and directories) and local references to the beginning of the CDH block.
- Data Descriptor: this structure is used only in a case when the LFH structure cannot be generated due to

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some reasons and some of the data required for LFH has become available after zipping the data. This structure is almost never used in practice

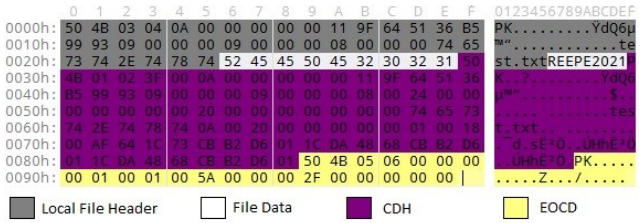

Fig.1. Internal contents of the ZIP archive in the HEX-editor

According to ISO/IEC 29500, OOXML-files represent ZIP-archives and various XML-files with specific structure; these files are combined in a single package. Such data fragmentation makes accessing data more simple and faster and minimizes the probability of damage. The [Content_Types].xml file located in the package root describes the types of document contents. A list of *.rels format document links is available at every package level. Basic text contents are located in document.xml/workbook.xml/ presentation.xml and have their own part of *.rels links containing links with other parts of xml content and URI for external links. References are saved only once and, therefore, can be easily modified when necessary. Figure 2 - 4 shows the structure of storing MS Word/Excel/PowerPoint data saved in ZIP format..

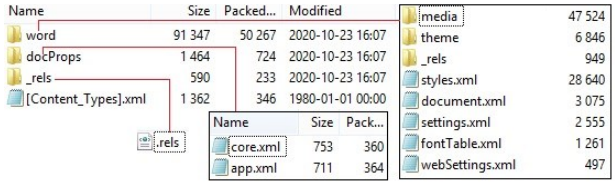

Fig.2. A structure of storing MS Word

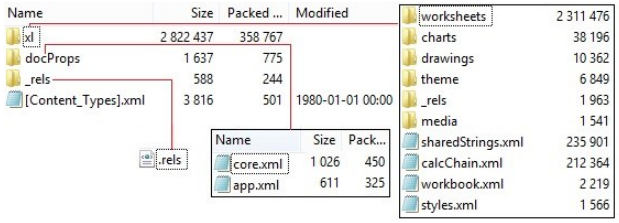

Fig.3. A structure of storing MS Excel

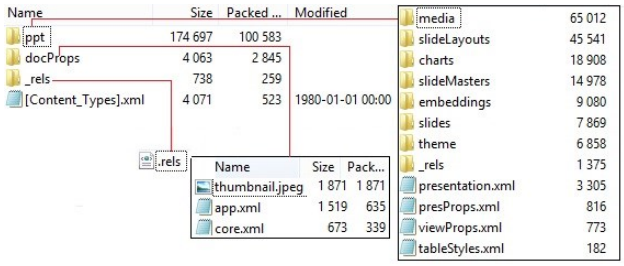

Fig.4. A structure of storing MS PowerPoint

III. Analysis of Information Hiding Type

MS Office-format documents have a very wide range of options and functions used when working with a document. With every release of a new MS Office version we obtain more and more options that simplify operations but, at the same time, make the document structure more and more complex.

In In this section, we would like to discuss several particularly important methods of hiding information. The analysis was conducted for each method, summarizing their strengths/weaknesses and addressing the stability of the stegocontainer, it's revealing and ease of implementation.

A. *Using Zip Format Comments*

Zip-file comments are optional information fields integrated into the Zip-file. They can be browsed, created, edited or deleted using the 7-Zip archive manager window. Comments are made both for the archive itself and individual components (files) at the archive level. In addition, each file has properties for attributes, where a comment field is also available, which can be used to introduce data.

Comment fields are used quite rarely and any information can result in suspected use of steganography. Comment fields are not subject to zipping and, therefore, can be read using any text editor. Figure 5-6 shows how the message was injected by HEX-editor (010 Editor) and easily detected.

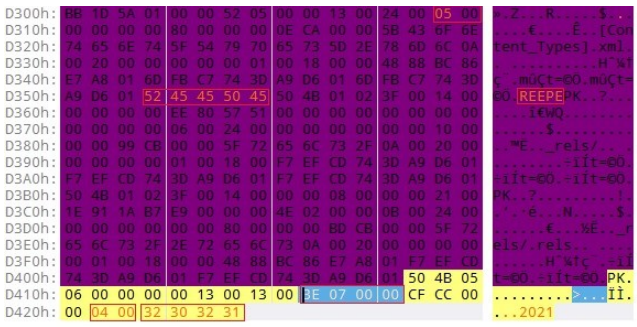

Fig.5. Adding comments by HEX-editor

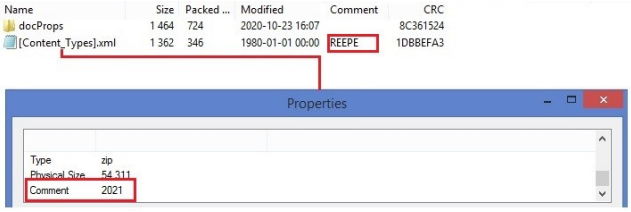

Fig.6. Detection of comments by 7-zip

Positive: an easy method for introducing data without using any additional software.

Negative: easy detection.

B. *Restricted Zip Fields*

CDH structure contains well-declared fields with their own properties and functions. A sequence of bytes is created in the archive where the values of corresponding fields related to this structure are stored. Figure 7 shows an OOXML file in a hexadecimal editor and the display of insert byte arrays with the message "Restricted zip fields" and edit values responsible for data size.

The "extra-field" field may be one of the fields that can be of interest for application; extrafield_length is the length of the special field for recording additional data attributes. It is almost never used and corresponds to zero. Comment fields are not subject to zipping and, therefore, can be read using any text editor.

Positive: well-defined structure fields are used, which can easily be addressed at any time and there is no need to look for the beginning of the stegocontainer. When using the 7-Zip archive manager for browsing, the text is not displayed.

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Negative: since the fields are rarely used, any information can result in suspected use of steganography. Data size checksum fields and so on must be recomputed to confirm the integrity of the archive.

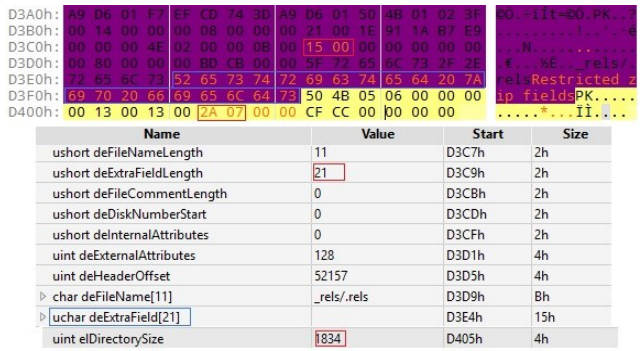

Fig.7. Add text "Restricted zip fields" in extrafield by Hex-editor

C. Embedding Files

OOXML documents provide the option to embed different media files and nested objects. Using media file inserts (drawingML object): using the functions developed by Microsoft, inserted images/videos are drawn on the document page together with text and do not need any additional processing. Embedded package: contains a complete package, which is internal or external to the reference structure.

For example, an embedded document can contain an image, an electronic table or a presentation document. In this case, it will not display the contents directly in the document. When the file is regarded as zip structure, information on the zipped file is recorded in LFH and CDH.

When deleting information on the file from LFH, this file will not be shown in the 7-Zip archive manager but it can successfully be retrieved from CDH. Such example is presented in figure 8-9. However, this method is very sensitive and unstable. When editing the document later, the CDH block will be updated and the undeclared file will disappear. To improve the stability of this method, it is suggested to consider adding data to the embedded document. For example, there are three images allocated in our document. The stegocontainer is divided proportionally to the image size and its separate portions are added to every image.

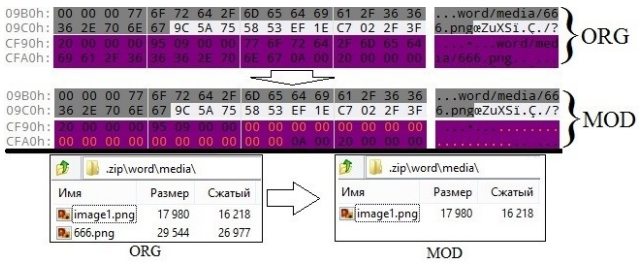

Fig.8. Delete a link to media file in the CDH block by HEX-editor

Positive: Medium revealing level. Revealing the stegocontainer will require time and experience. The stegocontainer size is larger compared to the previously discussed methods.

Negative: low stability in case of deleting LFH reference. Medium stability in case of adding media to files but this strongly depends on the availability of media files.

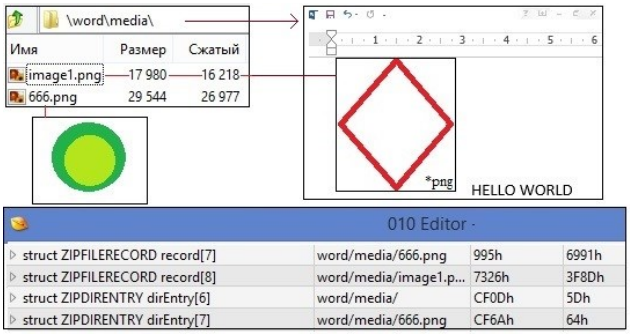

Fig.9. MS Word without link on 666.png

D. Additional Xml Fields

Since OOXML files are mainly available in XML format, which is suitable for introducing additional information due to its redundancy. This can involve adding text in between tag containers. Editing unused values based on the type of fonts/styles of text/subjects. Figure 10 shows an example of implementation, for this, the most voluminous data block was selected for a particular document in question «.xlsx/xl-sharedStrings.xml». In the selected part, 7621 places were counted between the containers of the "> <" tag. A 998 character message was embedded (converted to the bin) and recorded between tag containers.

Positive: Medium revealing level. Revealing the stegocontainer will require time and experience.

Negative: low stegocontainer capacity due to the small XML size. Breakdown of the stegocontainer when subsequently editing the document.

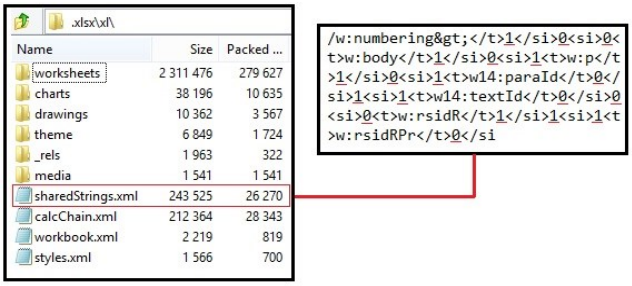

Fig.10. Embed a secret message in xml

E. Hidden Text

MS office functions enable the use of the option "hide text". The text remains a part of the document but is not displayed unless the inverse function is performed.

Using the characteristics of human vision, the message can be hidden by changing the text color according to the color of the background; however, the text itself will be displayed and occupy a certain space.

For this reason, it is more reliable to apply the LSB (Least Significant Bit) method to the color array. Characters representing a secret message in a normal text can be formatted in such a way that this will be invisible to an inexperienced text reader.

In In particular, the characters of a secret message can be highlighted in a different color that is slightly different from the color of other characters. An example of this method is presented in figure 11.

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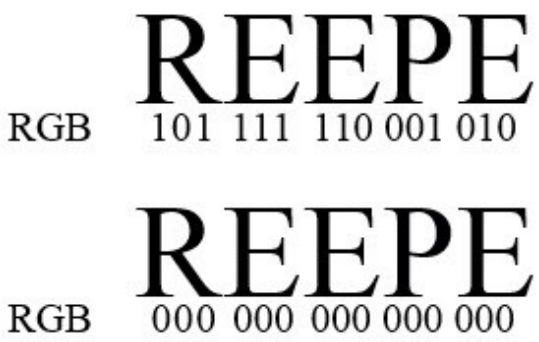

Fig.11. Example of embedding a message in a color

Positive: An inexperienced user will not notice hidden information.

Negative: It can easily be revealed by the internal MS Office inspector and change the color of the entire text.

F. Tracking Text

Tracking refers to a uniform change in the interval between letters (letter spacings). As opposed to kerning, where character spacing is regulated in certain pairs, tracking is applied to a group of symbols (word, line, paragraph, etc.). The OOXML provides the option to control letter and line spacings. Spaces are not always saved in XML. Similar to any XML, the space attribute must be specified in order to save spaces in OOXML. Invisible minor changes in values can be replaced using LSB. Figure 12 shows the value between characters "spacing w:val=100" and modified spacing w:val="-100".

Fig.12. Example of inter-character interval operation

The opposite combined option was also studied when the displacement was extremely significant, which allowed to display full words as a single character in the document and then change the font color to the background color. Figure 12 shows text displacement at spacing w:val="-260".

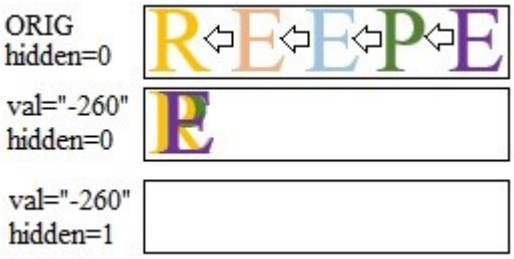

Fig.13. Example of combined use of inter-character distance and text color

Positive: When using LSB: Low level of revealing. The container capacity is proportional to the text size.

When using the combined approach: the MS inspector did not reveal the hidden text. The container capacity is unlimited.

Negative: When the stegocontainer is large, the document size will be increased respectively and arouse suspicions. An experienced analyst will reveal the combined option..

IV. Conclusion

OOXML format represents a compressed format of zip-files based on XML. Starting with Microsoft Office 2007, the default format is DOCX/ XLSX/ PPTX when creating a new document. The considered methods of hiding data do not violate the integrity and operability of documents, which allows us to consider them relevant.

In this article, we discussed the existing MS-Word file structure based on OOXML and analyzed what options could be applied in digital MS Office documents using the space XML/CDH/LFH/<spacing>. As a result, the performed research will allow us to develop criteria for revealing data loss.

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